Swaroop Nayak

Boston, MA | swaroopnayak1@gmail.com | (857)-426-8168 | LinkedIn: Swaroop Nayak | GitHub: swaroopnayak1 | swaroopnayak.com

Education

Northeastern University – *MS in Computer Science (GPA: 3.83/4.00)* — *Boston, MA*
Relevant Coursework: DBMS, Programming Design Paradigms, Algorithms, NLP — *Jan 2026 – Dec 2027 (Expected)*
Maharashtra Institute of Technology, Pune – *B.Tech CSE (GPA: 9.59/10.0)* — *July 2018 – June 2022*

Technical Skills

Languages: Python, Java, C#, C, C++, JavaScript, TypeScript
Cloud & DevOps: Microsoft Azure, Azure DevOps, Docker, Kubernetes, Git
Frameworks: .NET, Spring, React, Chart.js, Selenium, Playwright, SpecFlow, Cucumber, Gherkin, JUnit, NUnit, JEST
Databases: SQL Server, MongoDB, T-SQL
Monitoring & Debugging: Grafana, Prometheus
AI & ML: NLP, TensorFlow, PyTorch

Experience

Graduate Teaching Assistant — *Boston, MA*
Northeastern University — *Sept 2026 – Present*

- **Mentoring a 120+ student cohort** in **Programming Design Paradigms**, teaching **Java** and OOP design.
- **Assisting with grading** and providing feedback on code design, correctness, and testing.

SDET (Fullstack & Infra) - II — *Mumbai, MH*
Alter Domus — *Mar 2024 – Dec 2025*

- **Built the backend service layer (C#/.NET)** for an automation platform spanning **20+ microservices**, designing **REST API endpoints** and data models adopted by **20+ engineers**.
- **Engineered a distributed concurrency engine (C#)** to schedule and execute **100+ parallelized tasks** across **3 pipeline nodes**, reducing feedback loops by **40%**.
- **Designed containerized service environments** (**Docker/Kubernetes**) across **15 services** to ensure environment parity, cutting full-suite execution time from **24 hrs to 6 hrs (75% reduction)**.

SDET (Fullstack & Infra) - I — *May 2023 – Mar 2024*
Alter Domus

- **Optimized database throughput** by re-engineering data-access logic (**T-SQL stored procedures**, **MongoDB** queries) to batch-validate records across multiple tables, cutting validation time from **10 mins to under 1 min**.
- **Built a JavaScript/Chart.js dashboard** to visualize historical test-suite execution trends, and extended automation coverage to the frontend using **Jest** and **Selenium**.
- **Monitored service health** using **Grafana and Prometheus**, triaging **4–5 Sev 2–3 incidents weekly** and cutting time-to-resolve by rapidly isolating the failing service and root cause within **Kubernetes**.

Associate QC Engineer — *Bengaluru, KA*
Deloitte — *July 2022 – May 2023*

- **Built automated regression suites** using **Selenium with Java**, decreasing manual testing effort by **50%** for core modules.

Data Science and ML Intern — *Pune, MH*
Chatur Ullu Lab, MIT WPU — *Feb 2021 – July 2021*

- **Developed an AI classification model** using TensorFlow to identify **17 owl species**, achieving **85.3% accuracy** via **transfer learning (DCNNs, ImageNet)**.
- **Architected an ETL pipeline** using **BeautifulSoup4** to scrape and curate a bird-call audio dataset from xeno-canto.

Projects

Calendar Scheduling Service (Java, Spring)

- Designed a **Java/Spring calendar service (MVC)** supporting recurring event series (**TreeMap**-backed), scoped edit/delete (single/following/all), multi-calendar timezone conversion, and **ICS/CSV** export; applied **builder, strategy, command, and registry** patterns.
- Built a test suite with **95%+ coverage (JaCoCo)** and **95%+ mutation score (Pitest)**, plus **Selenium** regression coverage.

Amazon Product Review Summarizer (Python, PyTorch) Link

- Built an NLP pipeline using **LSTM (94.85% accuracy)** and a **BERT-based summarizer** to extract sentiment insights from reviews (ROUGE-1: 0.19).

Publications & Awards

Whose Hoot? Identification of Owl Species Using Call Recognition with Neural Networks – S. Nayak, S. Karande, P. Koparde (Jan 2022). SSRN

Innovator Award (Feb 2025) | **Innovator Award** (Feb 2024) | **Best Squad Award** (Dec 2023) – Alter Domus